

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Specification for Vent Stack

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R01	21-06-22	First Issue for Client's Review
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Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-004

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-005	Piping Class Specification
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-015	Specification for Piping
EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003	Process Flow Sheets 1 to 8
EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-007	Specification For Welding

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Abbreviations

Abbreviation	Definition
ASME	American Society for Mechanical Engineers
ASTM	American Society for Testing and Materials
API	American Petroleum Institute
BS	British Standard
BPVC	Boiler and Pressure Vessel Code
BVS	Block Valve Station
DEP	Design and Engineering Practice
DFT	Dry Film Thickness
DPR	Department of Petroleum Resources
FAT	Factory Acceptance Test
FEED	Front End Engineering Design
HAZ	Heat Affected Zone
MPI	Magnetic Particle Inspection
MS	Metering Station
NPS	Nominal Pipe Size
NDT/NDE	Non-Destructive Testing / Non-Destructive Examination
PS	Pigging Station
SAT	Site Acceptance Test
TGS	Terminal Gas Station
MESC	Material and Equipment Standard and code
MSS	Manufacturers Standardisation Society

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

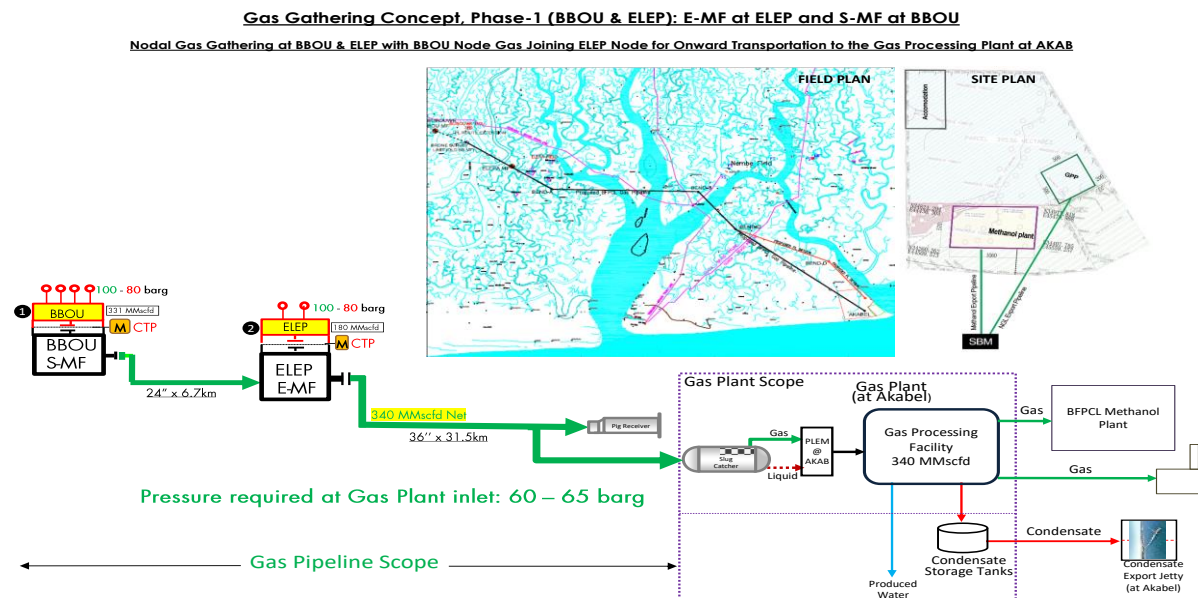


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification defines the minimum technical requirements for the supply of the Vent Stacks to be located at BBOU and Elepa. Manifold stations of the BFPCL Gas Gathering Pipeline Project - Phase 1. The vent system at Akabel pigging station shall be connected to the flare system of BFPCL plant at Akabel

This document shall be updated in Detailed engineering sizing of the vent stack is carried out.

1.3. Units of Measurement

In general, the SI system of units shall be used for the Project, with the following exceptions:

- Nozzle sizes shall be in inches.
- Flange ratings shall be in pounds.
- Pressure shall be in bar.
- Piping (NPS shall be in inches).
- Flow rate shall be in standard cubic feet per day.

1.4. Health, Safety and Environmental Requirements

VENDOR shall operate a formal Health, Safety and Environmental Management System, maintain awareness of the Health, Safety and Environmental (HS&E) legislation and regulations concerning safe and environmentally friendly working practices, and comply with these in all manufacturing, fabrication and testing operations pertaining to this specification.

OWNER's nominated representative shall have the right to carry out Health, Safety and Environmental audits at both VENDOR's and Sub-CONTRACTOR's works as he deems necessary to assess the effectiveness of VENDOR's HS&E management system.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1. Shell Specifications

Document number	Title
DEP 31.38.01.24	Piping – Engineering and Layout Requirements
DEP 31.22.00.31-Gen.	Unfired pressure vessels
DEP 30.10.60.18-Gen.	Welding of metals (amendments/supplements to API RP 582)

2.2. National and International Codes and Standards

Document Number	Description
AISC	Steel Construction Manual
API RP 520	Sizing, Selection, and Installation of Pressure-Relieving Devices in Refineries
API RP 521	Guide for Pressure-Relieving and Depressuring Systems
ASME II	Material Specifications
ASME V	Non-Destructive Examination
ASME VIII, DIV 1	Rules for Construction of Pressure Vessels
ASME IX	Welding
ASME B16.5	Pipe Flanges and Flanged fittings
ASME B16.47	Large Diameter Steel Flanges
ASME B16.9	Factory-Made Wrought Buttwelding Fittings
ASME B31.3	Process Piping

Document Number	Description
ASCE 7-05	Wind loads: Guide to the Wind Load Provisions of ASCE 7-05
AWS D1.1	Structural Welding Code
EN 10204	Metallic Products – Types of Inspection Documents.
EGASPIN	Environmental Guidelines and Standards for Petroleum Industry in Nigeria
EN 1991	Eurocode 1: Actions of Steel Structure
EN 1993	Eurocode 3: Design of Steel Structure
ISO 7531	Wire Rope slings for General purposes
ISO 9001	Quality Management Systems – Requirements
ISO 10474:1991	Steel and Steel Products – Inspection Documents
ISO 14001	Environmental Management Systems

3. Responsibilities and Scope Of Supply

3.1. Scope

Scope of work includes but not limited to design, materials procurement, fabrication, inspection and testing, surface preparation and painting, internal lining (if required), documentation, preparation for shipment, preservation and supply of Vent Stacks as specified in the datasheet, this specification and others listed herein.

3.2. Scope of Supply

The Vent Stacks shall be of the standpipe type and consist of two main components: The Vent Stack base structure and the Vent Stack extension. Acoustic silencers are not required unless it stated the relevant datasheet.

As a minimum, the supply for each Vent Stack shall include but not limited to the following:

- One structure body
- One man-way
- One vent gas inlet nozzle
- One vent gas outlet nozzle
- One drain nozzle
- One skirt and two earthing clips
- Two lifting lugs on the top section, for erection on the concrete foundation
- Bird screen

The Vent Stack extension shall also be a welded assembly, provided with two lifting lugs at the tip, for mounting on the Vent Stack base structure.

The scope of supply shall include the gaskets and bolting needed for connecting the base structure with the Vent Stack extension, as well as all necessary supporting structures (if required).

VENDOR shall supply detailed information for foundation design including

- Support details and calculations for equipment
- Support details and foundation loading diagram
- Lifting arrangement and lifting equipment drawings
- Design of anchor bolts

3.3. Scope of Services

VENDOR'S scope of services shall include but not be limited to:

- Fabrication, inspection and testing

- External and internal surface preparation, external painting and internal lining
- Spare parts for 2 years operation
- All certification and statutory approvals (if any).
- Preservation

3.4. **VENDOR Requirements and Deviations**

VENDOR shall make all possible efforts to comply strictly with the requirements of this specification, international codes and standards, datasheet, drawings and all other requirements as listed therein.

No deviation or exception from this specification shall be permitted without the written approval of the OWNER. Intended deviations shall be listed by VENDOR and supported by reasons for consideration by the OWNER. In the absence of any exceptions, it will be presumed that VENDOR fully complies with the specifications and attachments.

Compliance with this specification shall not relieve VENDOR of the responsibility of furnishing equipment and accessories of proper fabrication and workmanship to meet the applicable international codes & standards, client's requirements and other contractual obligations.

3.5. **Exclusions**

- On-site installation
- Commissioning
- Foundation design
- Supply of anchor bolts,

4. Design and Technical Requirements

4.1. Design

VENT STACKS and all other pressurized components shall be designed, fabricated, welded and tested in accordance with ASME BPVC Section VIII Division 1.

- The equipment shall be designed and manufactured to ensure low-maintenance, long-lasting and reliable operation.
- These stacks shall fully comply with DEP 31.22.00.31-Gen.Unfired pressure vessels, ME-SPC-0002 and other design requirements/documentations as listed in Section 2.1.
- The equipment shall have a design life of thirty (30) years.

4.2. Site Data

Equipment shall be designed for outdoor use in an onshore swampy atmosphere (low humid, non-corrosive and subject to high rainfalls.).

The environmental conditions for the site location are as stated in the Engineering Basis of Design – EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002.

4.3. Site and Environmental Conditions

The following site and environmental conditions have been extracted from Referencing paragraph 4.2.3 of Environmental and Social Impact Assessment Report prepared by Messrs. Fundamental Integrated Site Appraisal Services Limited.

4.3.1. Air Temperature

Average maximum (day time) temperature in the dry season months was about 32°C, and about 28°C in the rainy season. The minimum (night time) temperature level was fairly the same (23°C) through-out the year.

4.3.2. Weather

Two weather conditions are prevalent in the region:

Dry Season (Harmattan)

During Harmattan season (November – January, 2-3 months/year), the atmosphere is laden with extremely fine dust (particle size of 5 microns or less), which can penetrate equipment, block air filters, etc.

Wet Season

Heavy rainfall can be expected in the wet season which normally runs from the months of April through October. Lightning is very frequent especially during the rainy season and shall be specifically considered in the design of all relevant equipment.

4.3.3. *Rainfall*

Rainfall steadily increases from about 65mm in January to about 288mm in July after which it drops to about 245mm in August. This amount increases to a second maximum peak of about 331mm in September and then steadily drops to 47mm in December. The double maxima characteristics of Equatorial climate are very evident in the project area.

4.3.4. *Relative Humidity*

Parameters	Values
Mean	81%
Minimum	69%
Maximum	88%

4.3.5. *Wind Speed and Direction*

The monthly mean wind speed in the project area varies from 6.2km/h to 9.6km/h with a mean of 8.5km/h.

The south-westerly wind is predominant (about 60% of the annual winds) and prevails during the wet season (March - October). The north-easterly wind which predominate during the dry season (November– February) and also brings the harmattan accounts to about 32% of the annual winds.

4.3.6. *Seismic Zone*

The project area is classified as seismic zone 0.

4.4. Design Life

The Vent Stacks shall be designed based on a service life of 30 years.

4.5. Process Requirements

The nominative gas composition for the relevant fields where the vent stacks are to be installed are shown in the table below.

Gas Compositions from Nominated Fields

Field1: Bubouwe Bou Analogue		Field 2: Elepa	
Component	Mol.%	Component	Mol.%
N ₂	0.07	N ₂	0
CO ₂	0.59	CO ₂	2.22
C ₁	92.92	C ₁	81.66
C ₂	3.16	C ₂	6.13
C ₃	1.01	C ₃	2.59
iC ₄	0.61	iC ₄	0.75
nC ₄	0.44	nC ₄	0.81
iC ₅	0.25	iC ₅	0.45
nC ₅	0.19	nC ₅	0.33
C ₆	0.23	C ₆	0.65
C ₇	0.19	C ₇	0.9
C ₈	0.04	C ₈	0.87
C ₉	0.03	C ₉	0.38
C ₁₀	0.02	C ₁₀	0.3
C ₁₁	0.03	C ₁₁	0.23
C ₁₂₊	0.22	C ₁₂	0.22
C ₁₃	NA	C ₁₃	0.23
C ₁₄	NA	C ₁₄	0.22
C ₁₅	NA	C ₁₅	0.19
C ₁₆	NA	C ₁₆	0.15
C ₁₇	NA	C ₁₇	0.12
C ₁₈	NA	C ₁₈	0.11
C ₁₉	NA	C ₁₉	0.06
C ₂₀₊	NA	C ₂₀₊	0.43
TOTAL	100	TOTAL	100
Flash CGR	7	Flash CGR	59.7

4.6. General Design Principles

The gaseous hydrocarbon streams exiting the Vent Stack will form mixtures in the flammable range, as the gas mixes with air. Dilution of the hydrocarbon gas stream to below its lower flammable limit is desirable and achievable. For this purpose, the relief stream shall leave the tip of the Vent Stack with a certain velocity to entrain additional ambient air by means of jet momentum forces generated by the gas stream released.

API Recommended Practice 520 considers the probability of the simultaneous occurrence of lightning and emergency depressurisation of parts of the entire station as negligible. Intermittent, low-momentum discharges at low-velocity, however, may increase the probability of lightning ignition.

Since depressurisation of the entire station or certain sections of the pipeline is typically infrequent and of short duration only, no particular precautionary measures shall be taken to attenuate the noise so generated; the operating personnel is expected to wear protective ear sets.

4.7. Vent Stack Base Structure

4.7.1. Structure Components

The Vent Stack base structure shall be a pressure vessel designed for an internal pressure according to the Vent Stack's datasheet. The applicable design code shall be the latest edition of the ASME Boiler and Pressure Vessel Code Section VIII, Division 1.

The vent gas inlet nozzle, vent gas outlet nozzle and the drain nozzle (where applicable) shall be in compliance with Specification for Piping Material Class and Specification for Piping Material referenced in section 2.1 herein.

The structure body (shell, bottom closure and transition piece) shall be composed of low temperature material as specified in the datasheet.

Details of the structure components (including sizes, height etc.) shall be specified in the datasheet.

4.7.2. Structural Body

The body of the Vent Stack base structure shall be a welded structure, composed of three components:

The body top shall be of the concentric reducer type to provide the transition from the large-size body shell to the small-size vent gas outlet nozzle.

The body shell shall be of cylindrical shape, sufficiently long to accommodate both the vent gas inlet nozzle, the man-way and the condensate sump. The vent gas inlet nozzle and the man-way shall be apart by 90 degrees. The body shell shall externally be reinforced at the location of both

the vent gas inlet nozzle and the man-way.

- The body bottom shall be a cap to ASME B16.9.

The Vent Stack shall be connected to the station's grounding system.

The Vent Stack shall be properly fixed on a concrete foundation that is sufficiently sized to accommodate both the maximum thrust developed during blow down activities and the maximum overturning moment induced by the design wind velocity.

4.7.3. Vent Gas Inlet Nozzle

The size of the vent gas inlet nozzle shall be identical with the size of the related vent gas line as shown on the data sheet and shall be arranged horizontally at the structure body. The distance between the nozzle's centreline and the bottom surface of the skirt flange shall be shown on the datasheet.

The vent gas inlet nozzle shall be flanged to ASME B16.5 at its exterior side and extended into the interior of the body structure, as shown in the datasheet. Part of the nozzle's interior side shall be sloped downward (by approximately 30 degrees) to direct the vent gas stream toward the bottom of the Vent Stack base structure.

4.7.4. Vent Gas Outlet Nozzle

The size of the vent gas outlet nozzle shall be identical with the size of the vent gas inlet nozzle. The nozzle itself shall be arranged axially at the top of the structure body and shall be a welded structure consisting of a sufficiently long nozzle neck and a flange in accordance with applicable provisions of the ASME Boiler and Pressure Vessel Code. The nozzle neck may be either a seamless or longitudinally welded piece of pipe. The nozzle components shall be made from low-temperature material.

4.7.5. Drain Nozzle

The structure body shall be provided with a drain, it shall be a flanged drain connection of size not less than NPS 2. The drain connection shall be attached axially at the centre of the structure body's bottom closure and continue horizontally to the outside of the skirt where it shall terminate with a flange. The materials of the drain connection shall be in accordance with Specification for Piping Material Class and Specification for Piping Material referenced in section 2.1 herein.

4.7.6. Man-way

Access to the interior of the Vent Stack shall be provided by means of a NPS 24 man-way, consisting of a nozzle, a cover plate, and a davit for easy handling of the cover plate. The man-way shall be attached to the shell of the structure body by welding. The hole to be cut into the body shell shall be reinforced externally. The man-way shall be complete with gasket, bolts and nuts.

The nozzle (consisting of nozzle neck and nozzle flange) and the cover plate shall be designed and sized in accordance with applicable provisions contained in the ASME Boiler and Pressure Vessel Code. The material for these components shall be low temperature steel.

4.7.7. Skirt

The structure body shall rest on a vertical skirt, consisting of a cylindrical shell and a ring-type foundation flange. The height of the skirt shall be according to the relevant datasheet. The material for the skirt shell shall be the same as used for the bottom cap of the structure body.

The skirt shall be capable without any sign of permanent deformation of transferring to the foundation the overturning moment that the Vent Stack is subjected to when exposed to the design wind speed. This applies particularly to the foundation flange with regard to thickness and size as well as number of anchor holes. The foundation flange shall be supported against the skirt shell, by means of triangularly shaped ribs that are to be arranged between the each anchor hole.

The line of skirt attachment to the structure body shall be the mean diameter of the skirt shell and the mean diameter of the structure body's shell.

The skirt shall be attached to the bottom closure of the base structure body by means of continuous fillet weld. The face of the skirt shell that is to be joined with the bottom closure shall be horizontal. The leg of the welding seam at the skirt shell shall be equal to the thickness of the shell and the other leg shall be twice the thickness of the skirt shell.

All welding on the skirt shall be in accordance with AWS D1.1.

4.7.8. Earthing Clips

One earthing clip shall be provided at the skirt, approximately 500 mm above the bottom surface of the skirt's foundation flange.

4.7.9. Lifting Lugs

The concentric reducer section of the structure body shall be provided with two lifting lugs for vertical lifting of the complete Vent Stack base structure. The lifting lugs shall be arranged 180 degrees apart, and each lug shall be capable to carry individually the weight of the complete Vent Stack base structure.

4.8. Vent Stack Extension

The Vent Stack shall consist of a piece of pipe (seamless or longitudinally welded) and a flange for connection to the vent gas outlet nozzle of the base structure. The size of the Vent Stack shall be identical with the size of the vent gas inlet nozzle.

The material for both components shall be low-temperature carbon steel. The flange shall be designed and sized in accordance with applicable provisions contained in the ASME B16.5. The

length of the Vent Stack shall be selected according to the process and blow down calculations.

The Vent Stack shall be provided with two lifting lugs for vertical lifting. The lifting lugs shall be arranged at the tip of the Vent Stack base, 180 degrees apart. Each lug shall be capable of carrying individually the weight of the Vent Stack.

The Vent Stack tip shall be equipped with rain deflector with bird screen. Refer to the relevant PEFS referenced in section 2.1 herein.

To reduce vibrations at the tip of the Vent Stack extension, the Vent Stack extension can be equipped with a vibration damper. VENDOR to accommodate this in the design calculation.

4.9. Calculation and Stress Analysis

The calculations and stress analysis to be carried out by the VENDOR shall demonstrate that the self-supporting Vent Stack will meet the requirements of design code for the worst case of operation.

5. Fabrication

5.1. General

The Vent Stack base structure and the Vent Stack extension shall be shop-welded assemblies fabricated in compliance with applicable provisions of the ASME Boiler and Pressure Vessel Code (BPVC) Section VIII division 1.

Vendor shall also comply with DEP 31.22.00.31-Gen.Unfired pressure vessels for details and requirements for Fabrication.

Welding of the Vent Stack base structure shall be done on the basis of approved welding procedures and carried by welders that are qualified in compliance with the ASME BPVC. The welding seams of the structure body shall have full penetration and complete fusion.

VENDOR shall submit the weld procedures, welding details and welding procedure qualification records to OWNER for review and approval, prior to the commencement of any welding. Radiographic, magnetic particle and dye penetrant inspection shall be performed by experts having an internationally recognized qualification.

5.2. Shop Inspection and Testing

5.2.1. General

The VENDOR shall ensure that the quality plans for inspection and testing are complying with the specific inspection, testing and certification requirements established in this specification and the referenced codes as well as standards.

The VENDOR shall provide a quality plan for the fabrication, inspection and testing of the Vent Stack, for review and approval by OWNER.

The VENDOR shall be responsible for the inspection and testing of the Vent Stack. If there are Hold points to be observed, they shall be included in the Inspection and Testing Plan. Shop-inspection and shop testing or witnessing of the shop-inspection and the shop testing by Third Party Inspection shall not relieve the responsibility of the VENDOR.

After the commencement of the design, material procurement, fabrication or testing of the Vent Stack, OWNER designated representatives shall have free access to VENDOR's workshop for inspection of the progressing work on the Vent Stack. All necessary certification of materials, shop test data, etc. shall be made available to the OWNER designated representatives.

Upon completion of the examinations and the hydrostatic tests, the VENDOR shall submit examination and test reports, certifying that the Vent Stack comply with the specified requirements. The reports shall be approved by OWNER.

5.2.2. Testing

The structure body of the completely assembled Vent Stack base structure and the Vent Stack extension shall be subjected to a hydrostatic test (specified in the datasheet). OWNER may send their representative or third-party inspection agency to witness the hydrostatic tests that are to be carried out by the VENDOR in the manufacturer's workshop.

OWNER shall be informed of the date of the hydrostatic shop test(s) at least four weeks prior to the scheduled date – for approval and decision on witnessing.

5.2.3. Examination

All longitudinal and circumferential welding seams of a Vent Stack (except that of the skirt) shall be examined at full length by radiography. The welding seams connecting the skirt foundation flange with the skirt shell, vent gas outlet flange with the corresponding nozzle neck, and the Vent Stack flange with the Vent Stack pipe shall be examined by non-destructive testing, either the magnet particle method or the dye penetrant method.

The examination to be performed after any post-weld heat treatment and the acceptance criteria shall be in accordance with the requirements of ASME Section V.

5.2.4. Material Control Records

The VENDOR shall submit details of material control recording procedure to OWNER for review, prior to commencement of the fabrication. The VENDOR shall submit details of his traceability procedure for the OWNER's review and approval.

5.2.5. Material Certification

VENDOR shall ensure material certifications meet the requirements as stated EN 10204.

Dimension check shall be recorded in the report and certified at least by means of an acceptance test certificate 3.1 in accordance with EN10204. The pressure test shall be certified on acceptance certificate 3.2 in accordance with EN 10204. Material test certificates TYPE 3.1 for wetted parts and TYPE 2.2 for structural parts.

6. Name Plate Marking

VENT STACKS nameplate shall be permanently mounted on a dedicated bracket sufficiently strong to allow nameplate stamping.

The nameplate shall be located in a clearly visible place on the shell.

Refer to DEP 31.22.00.31-Gen.Unfired pressure vessels section 5.17 for requirements for nameplate marking.

7. Documentation

In addition to the requirement of DEP 31.22.00.31-Gen.Unfired pressure vessels, the following documentations shall be submitted by VENDOR as required:

- Material certifications according to EN 10204
- Report of heat treatment, including diagrams
- FAT procedures and report
- SAT procedures
- Commissioning procedures
- Lifting study
- Acceptance Certificate (issued by Notified Body) of pressure test and final inspection
- Installation drawings
- All applicable welding certificates and records
- All applicable NDT certificates and records
- All applicable destructive test specifications, procedures, certificates and records
- Documentation of all carried out repair works
- Documentation of all waivers and deviations from specifications
- Coating and lining inspection report
- Flushing, Cleaning and Drying report
- Spare parts and materials for commissioning, start-up, and 2 years operation
- Certificates of gas-tight connections

8. Cleaning, Painting and Shipping Protection

8.1. Drying and Cleaning

On completion of inspection and testing, the major components of the Vent Stacks shall internally and externally be thoroughly cleaned and dried, prior to preparation for shipment.

8.2. Painting and Coating

The Vent Stacks shall be painted externally and internally, in accordance with the Painting and Coating Specification of the project.

The paint shall only be applied after testing and acceptance of the Vent Stack. The surfaces to be painted shall be mechanically cleaned to SA 2 ½. The paint systems to be applied on the mechanically cleaned surfaces shall be as detailed hereinafter.

All exterior surfaces of the Vent Stack (such as Vent Stack, Vent Stack base structure, nozzles, man-way, inner side of the skirt etc.) shall receive the following paint system. The total dry film thickness (DFT) shall be at least 290 µm.

Primer Zinc-epoxy DFT = 80 µm

Intermediate coat Epoxy high-solid 2-comp. Polyurethane DFT = 160 µm

Top coat Epoxy high-solid 2-comp. Polyurethane DFT = 50 µm

All interior surfaces of the Vent Stack (such as Vent Stack, body of the base structure, nozzles, man-way nozzle and man-way cover plate) shall receive the following paint system. The total dry film thickness (DFT) shall be at least 175 µm.

Primer Zinc-ethyl silicate DFT = 75 µm

Intermediate coat Silicone acrylic DFT = 50 µm

Top coat Silicone acrylic DFT = 50 µm

8.3. Shipment

VENDOR shall take all necessary precautions in loading, blocking, and bracing the Vent Stack to prevent damage during shipment.

The VENDOR's own proposal for the preparation for shipment may be reviewed as an alternative but shall be subject to review and approval by OWNER.

The Vent Stack base structure and the Vent Stack shall be packed separately and seaworthy, in a way that prevents damage during shipment and storage at site. Adoption of a suitable packing method (e.g. wooden boxes) is at the sole discretion of the VENDOR. Should any loss, damage

and / or deterioration be caused due to improper packing, the consequences shall be the responsibility of the VENDOR.

Flanges shall be sealed air-tight by covering them with shipping gaskets and metallic caps that are to be secured on the flanges by shipping bolting.

Vent Stack's components (Vent Stack base structure and Vent Stack) shall be shipped in wooden boxes, as per specific design to be done by the VENDOR.

The wooden boxes shall be clearly marked with the content and tag number. The marking shall be waterproof and tear-resistant and shall be attached with stainless steel wire to the wooden boxes.

9. Quality Assurance

9.1. Inspection Authorization

Examination or tests may be reviewed or witnessed by OWNER or its authorized agent at VENDOR's facility. A general inspection of the facility for quality control procedures may be made as selected by the representative. Repairs shall also be inspected as selected by OWNER's representative after the repairs have been completed.

9.2. Major Repairs

Major repairs shall not be performed without authorization. Welding procedures may be furnished for evaluation by OWNER as an alternative.

9.3. Performance Guarantee

VENDOR shall have overall responsibility for the proper fabrication workmanship & satisfactory functioning of the VENT STACKs and is required to guarantee the performance of his scope of supply as defined in this specification, data sheets and other documentation referenced herein.